NBA HIKING PIPELINE: CORRELATING BASKETBALL STATS WITH TRAIL DIFFICULTY

5IF INSA - Foundations of Data Engineering Project

AUTHORS

- Sebastian Pica
- Luis Schwarz
- Shuyan Dou

INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON

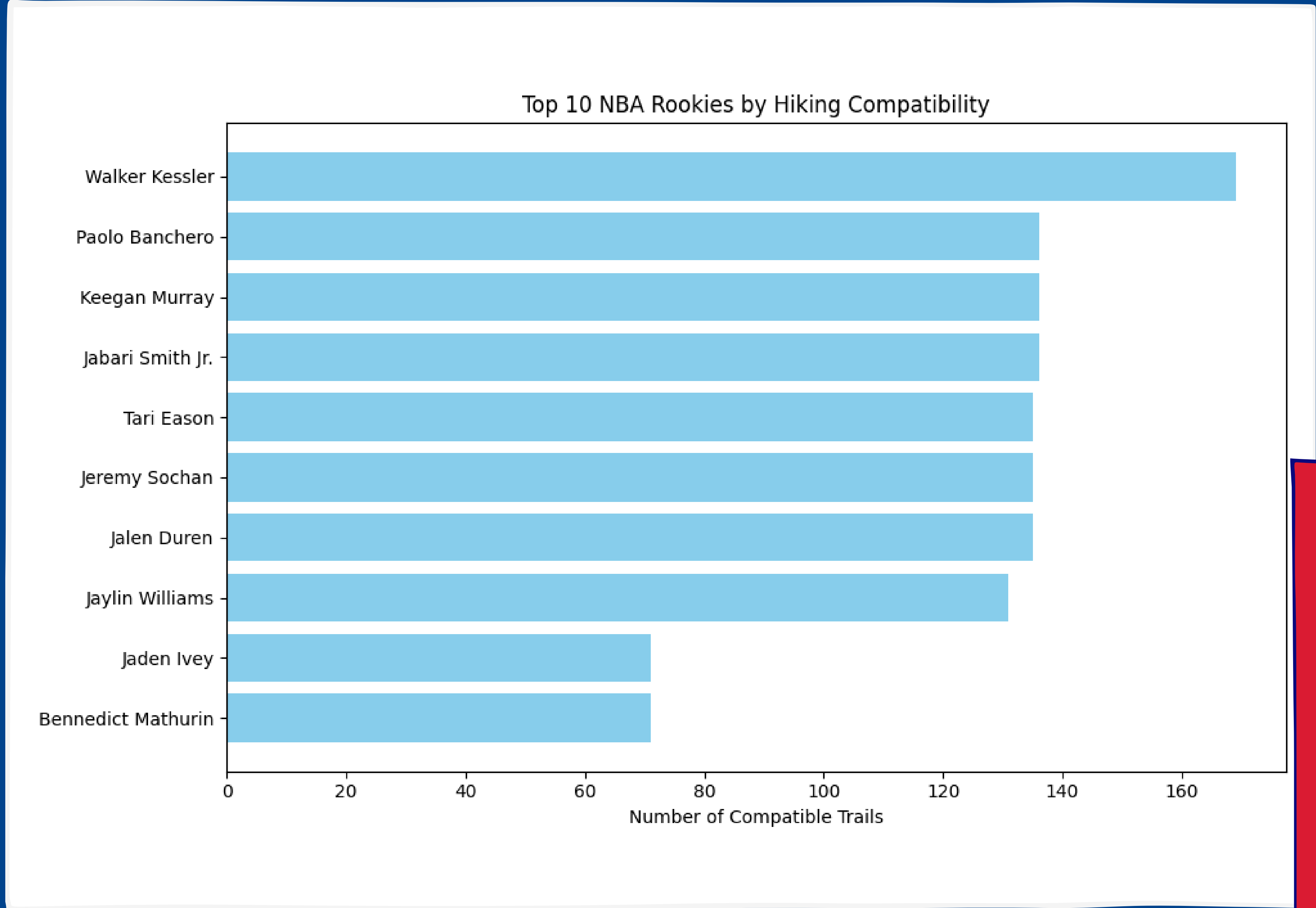
01. INTRODUCTION

This project combines two interests of our team members: Basketball and Hiking. We built a data pipeline to determine which NBA players are physically best suited for hiking trails in 'The Gorge' by mapping athletic stats to trail requirements.

02. METHODOLOGY

We engineered three compatibility metrics:

- Endurance: Minutes per Game× Games Played vs. Trail Distance.
- Strength: Rebounds + Blocks vs. Elevation Gain.
- Agility: Steals + Assists vs. Trail Hazards.



03. DATA SOURCES:

The data we work with is stored in csv which we obtained from nba stat tracking websites and open source data stores:

- NBA Stats: Player performance metrics (CSV).
- Hiking Trails: Distance & Elevation metadata (CSV).
- Trail Hazards: Safety flags & danger ratings (CSV).

04. GENERAL RESULTS

We implemented two DAGs to analyze the general NBA population and team fitness:

- Top 15 Hikers: Players with high durability (High Games Played and Minutes) dominated the rankings as they qualify for the most trails.
- Top 5 Teams: Aggregating scores by team identified "outdoorsy" rosters. Teams with deep benches and high average fitness ranked highest.

We have also created a dedicated pipeline to analyze the 2022-23 Rookie class hiking ability. This required us to further populate the data set with that season's rookies.

05. ARCHITECTURE & TOOLS

The project relies on a containerized ELT (Extract, Load, Transform) architecture for reproducibility:

- Docker: Encapsulates the environment (Airflow, Postgres, Volume mounts).
- Airflow: Orchestrates 3 parallel DAGs to manage dependencies.
- PostgreSQL: Stores the Staging (Raw) and Gold (Final) data layers.
- Pandas & Matplotlib: Handles data cleaning (Regex), transformation, and reporting

06. CONCLUSION

This project successfully correlates disparate domains like Sports and Nature using Data Engineering principles.

- Success: Automated ingestion and cleaning of messy real-world text data.
- Challenge: Parsing inconsistent units (e.g., "1,200 ft" vs "1200") required robust Regex patterns.
- Future: Possible implementation of a dynamic web frontend for interactive filtering.